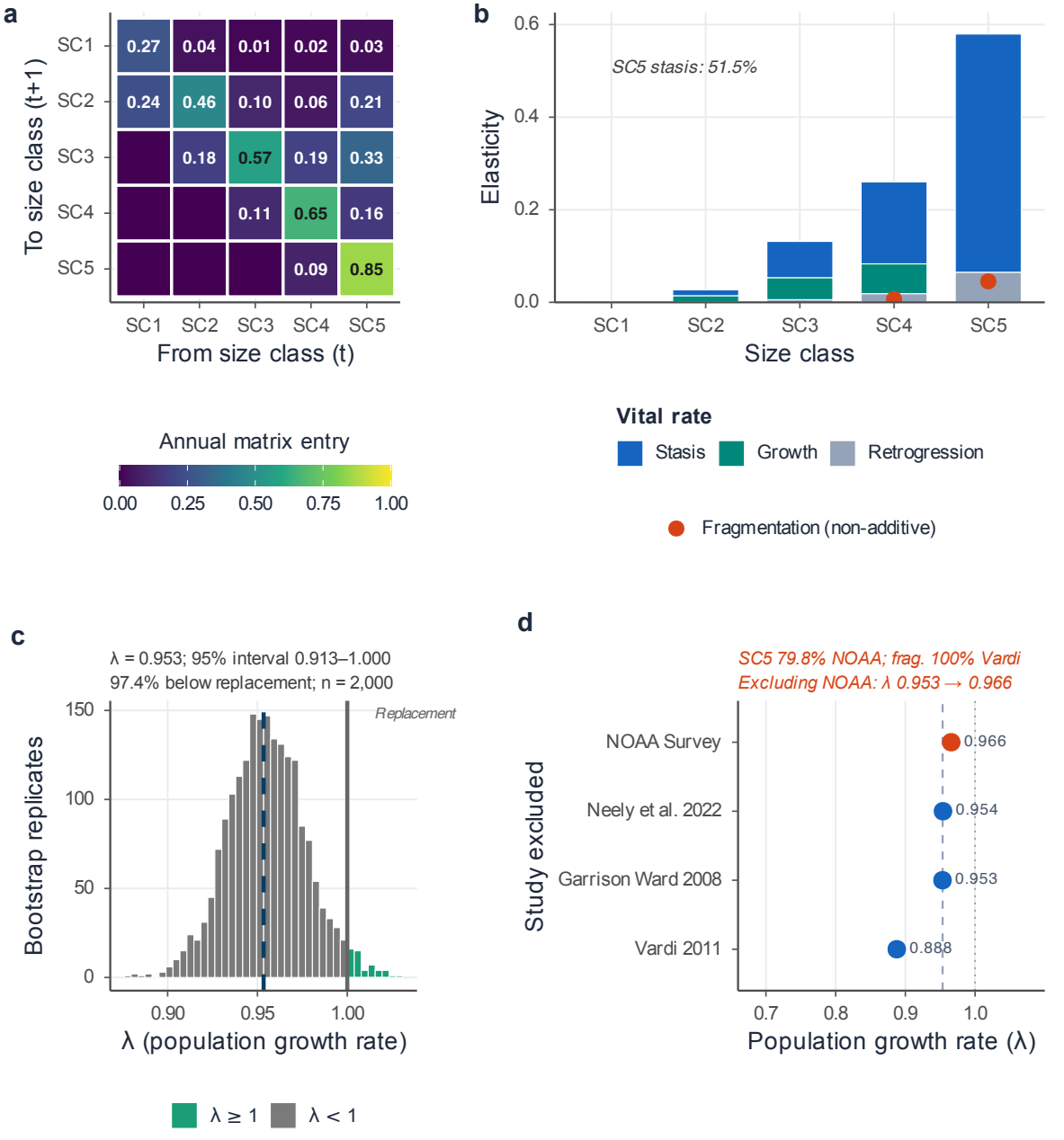

a
SC1
SC2
SC3
SC4
SC5
0.27
0.04
0.01
0.02
0.03
0.24
0.46
0.10
0.06
0.21
0.18
0.57
0.19
0.33
0.11
0.65
0.16
0.09
0.85
To size class (t+1)
From size class (t)
Annual matrix entry
0.00
0.25
0.50
0.75
1.00
b
SC5 stasis: 51.5%
Elasticity
0.6
0.4
0.2
0.0
Size class
Vital rate
Stasis
Growth
Retrogression
Fragmentation (non-additive)
c
λ = 0.953; 95% interval 0.913–1.000
97.4% below replacement; n = 2,000
Replacement
Bootstrap replicates
150
100
50
0
0.90
0.95
1.00
λ (population growth rate)
λ ≥ 1
λ < 1
d
SC5 79.8% NOAA; frag. 100% Vardi
Excluding NOAA: λ 0.953 → 0.966
Study excluded
NOAA Survey
Neely et al. 2022
Garrison Ward 2008
Vardi 2011
0.966
0.954
0.953
0.888
0.7
0.8
0.9
1.0
Population growth rate (λ)